Year 13 Period 10 Work 21

Q1.

A circle *C* has centre (−1, 7) and passes through the point (0, 0). Find an equation for *C*.

(4)

$r = \sqrt{(0-7)^2 + (0--1)^2}$

$r = 5\sqrt{2}$

$\therefore$

$(x+1)^2 + (y-7)^2 = (5\sqrt{2})^2$

(Total 4 marks)

Q2.

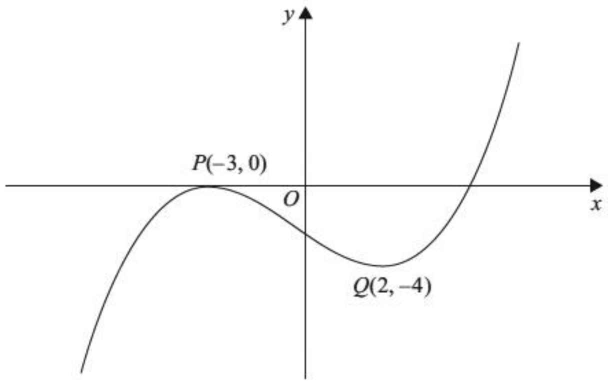


Figure 1

Figure 1 shows the graph of equation $y = \mathrm{f}(x)$.

The points P (− 3, 0) and Q (2, − 4) are stationary points on the graph.

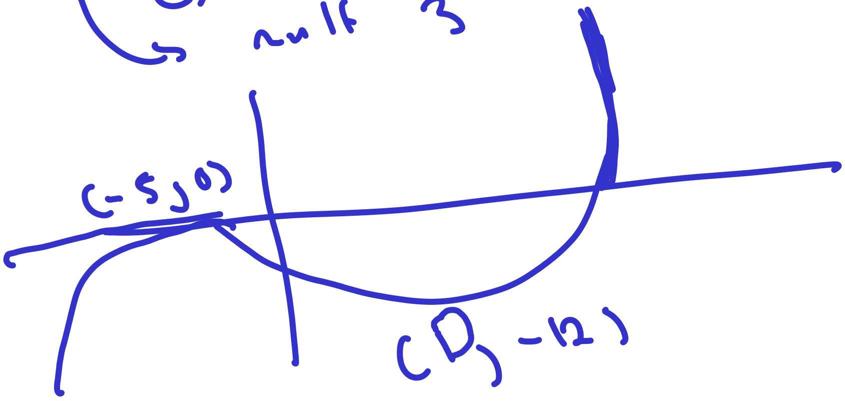
Sketch, on separate diagrams, the graphs of

(a) $y = 3\mathrm{f}(x + 2)$

↳ −2 from x

↳ mult 3

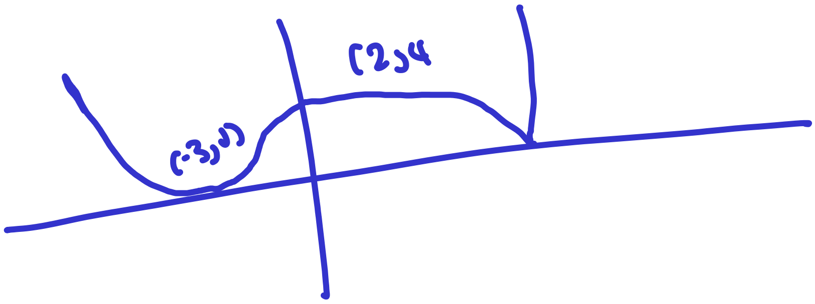
(3)



(b) $y = |\mathrm{f}(x)|$

↳ all values of y would be postive

(3)



On each diagram, show the coordinates of any stationary points.

(Total 6 marks)

Q3.

Jess started work 20 years ago. In year 1 her annual salary was £17 000. Her annual salary increased by £1 500 each year, so that her annual salary in year 2 was £18 500, in year 3 it was £20 000 and so on, forming an arithmetic sequence. This continued until she reached her maximum annual salary of £32 000 in year *k*. Her annual salary then remained at £32 000.

(a) Find the value of the constant *k*.

$a = 17{,}000$ $\therefore$ $U_n = a + (n-1)d$

$d = 1{,}500$

(2)

$32{,}000 = 17{,}000 + (n-1)\,1500$

$15000 = (n-1)\,1500$

$10 = (n-1)$ $n = 11$//

(b) Calculate the total amount that Jess has earned in the 20 years.

$a = 17{,}000$ $n = 20$

$d = 1{,}500$

(5) 1 ✗

$S_n = \frac{n}{2}(2a + (n-1)d)$

$S_n = \frac{20}{2}(2(17{,}000) + (20-1)\,1500)$

£625,000//

↪ forgot to realise that it wont increase past $n = 11$//

(Total for question = 7 marks)

Q4.

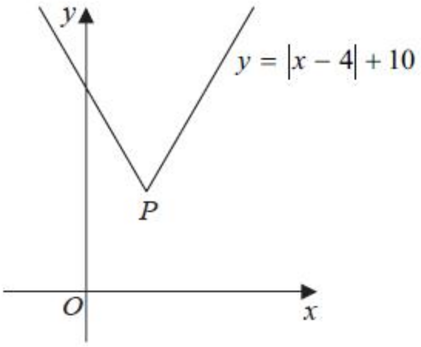


Figure 1

Figure 1 shows a sketch of the graph with equation $y = |x - 4| + 10$

The point *P*, shown in Figure 1, is the vertex of the graph.

(a) State the coordinates of *P*

P is the min ∴ $x = 4$

a ∴ $y = 10$ as $|4 - 4| = 0$

$(4, 10)$

(2)

(b) Use algebra to solve

$$|x - 4| + 10 = 3x$$

(Solutions relying on calculator technology are not acceptable.)

(2)

$|x - 4| + 10 = 3x$

$(x - 4) + 10 = 3x$

$x + 6 = 3x$

$6 = 2x$

$x = 3$

3 doesn't work

$-(x - 4) + 10 = 3x$

$-x + 4 + 10 = 3x$

$-x + 14 = 3x$

$14 = 4x$

$x = \frac{7}{2}$

$x = \frac{7}{2}$ / alvae /

(Total for question = 4 marks)

Q5.

(a) Express 2 sin θ − 4 cos θ in the form R sin(θ − α), where R and α are constants, $R > 0$ and $0 < \alpha < \frac{\pi}{2}$
Give the value of α to 3 decimal places.

$r = \sqrt{2^2 + 4^2}$

$r = 2\sqrt{5} \rightarrow \sqrt{20}$

$\theta = \tan^{-1}\left(\frac{4}{2}\right)$

$a = 1.107$

$R = \sqrt{20}\sin(\theta - 1.107)$ //

(3)

$$H(\theta) = 4 + 5(2\sin 3\theta - 4\cos 3\theta)^2$$

Find

(b) (i) the maximum value of H(θ),

(ii) the smallest value of θ, for $0 \le \theta < \pi$, at which this maximum value occurs.

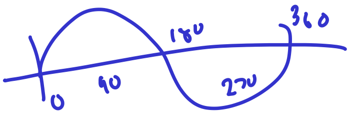
(3)

$H(\theta) = 4 + 5(\sqrt{20}\sin(\theta - 1.107))^2$

the max is when $\sin(\theta - 1.107) = 1$

$4 + 5(\sqrt{20})^2$

$= 104$ //



$\sin(3\theta - 1.107) = 1$

$0 \le \theta < \pi$

$0 \le 3\theta \le 3\pi$

$-1.107 \le 3\theta - 1.107 \le 3\pi$

$3\theta - 1.107 = 1.570796\ldots, \frac{5}{2}\pi$

$3\theta = 0.84, 2.99$

$\theta = 0.89$ //

Find

(c) (i) the minimum value of H(θ),

(ii) the largest value of θ, for $0 \le \theta < \pi$, at which this minimum value occurs.

(3)

when $\sin(\theta - 1.107) = 0$

$4 + 5(0)^2 = 4$ /

$\sin(\theta - 1.107) = 0$

$3\theta - 1.107 = 2\pi$

$\theta = 2.416358$ //

(Total 9 marks)

Q6.

(a) Find the binomial expansion of

$$\sqrt[3]{(8-9x)}, \qquad |x| < \frac{8}{9}$$

in ascending powers of x, up to and including the term in x^3. Give each coefficient as a simplified fraction.

(6)

$(8 - 9x)^{1/3} \rightarrow 8^{1/3}[1 - \frac{9}{8}x]$

$\hookrightarrow 2[1 - \frac{9}{8}x]^{1/3}$

$2[1 + (-\frac{9}{8}x)(\frac{1}{3}) + \frac{\frac{1}{3}(\frac{1}{3}-1)}{2!}(-\frac{9}{8}x)^2 +$

$\frac{\frac{1}{3}(\frac{1}{3}-1)(\frac{1}{3}-2)}{3!}(-\frac{9}{8}x)^3]$

$2[1 - \frac{3}{8}x - \frac{9}{64}x^2 - \frac{45}{512}x^3)$

$2 - \frac{3}{4}x - \frac{9}{32}x^2 - \frac{45}{256}x^3$

(b) Use your expansion to estimate an approximate value for $\sqrt[3]{7100}$, giving your answer to 4 decimal places. State the value of x, which you use in your expansion, and show all your working.

~~**(3)**~~

0

(Total 9 marks)

Q7.

Ben is studying the Daily Total Rainfall, x mm, in Leeming for 1987

He used all the data from the large data set and summarised the information in the following table.

x	0	0.1–0.5	0.6–1.0	1.1–1.9	2.0–4.0	4.1–6.9	7.0–12.0	12.1–20.9	21.0–32.0	tr
Frequency	55	18	18	21	17	9	9	6	2	29

(a) Explain how the data will need to be cleaned before Ben can start to calculate statistics such as the mean and standard deviation.

the tr data is between 0 and 0.25

∴ Ben could set it to 0

(2)

Using all 184 of these values, Ben estimates $\sum x = 390$ and $\sum x^2 = 4336$

(b) Calculate estimates for

(i) the mean Daily Total Rainfall,
(ii) the standard deviation of the Daily Total Rainfall.

(3)

bi $\frac{\Sigma x}{n} = \frac{390}{184} = 2.12$ 3sf

bi $\sqrt{-\left(\frac{\Sigma x}{n}\right)^2 + \left(\frac{\Sigma x^2}{n}\right)} = 4.37$ 3sf //

Ben suggests using the statistic calculated in part (b)(i) to estimate the annual mean Daily Total Rainfall in Leeming for 1987

(c) Using your knowledge of the large data set,

(i) give a reason why these data would not be suitable,
(ii) state, giving a reason, how you would expect the estimate in part (b)(i) to differ from the actual annual mean Daily Total Rainfall in Leeming for 1987

(2)

(Total for question = 7 marks)

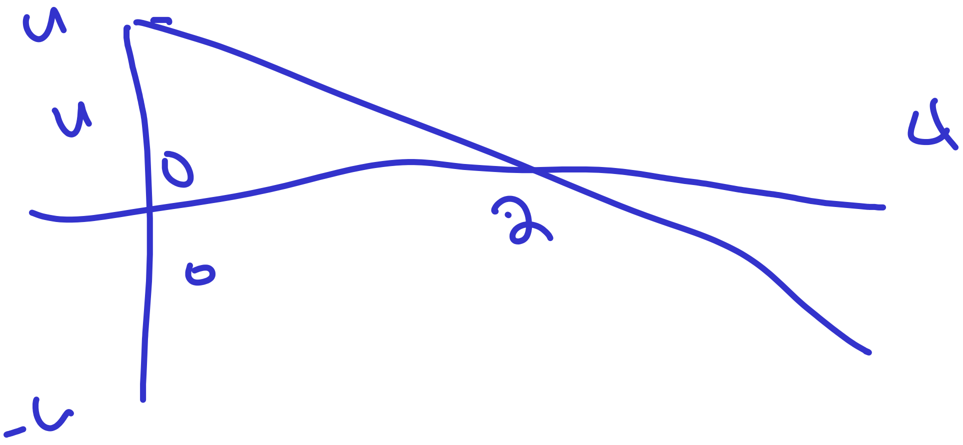
u↑ then ↓u

Q8.

A small ball is projected vertically upwards from ground level with speed u m s^{-1}. The ball takes 4 s to return to ground level.

(a) Draw, in the space below, a velocity-time graph to represent the motion of the ball during the first 4 s.

(2)



(b) The maximum height of the ball above the ground during the first 4 s is 19.6 m. Find the value of u.

(3)

$\frac{1}{2} + 2t \quad u = 19.6$

$u = 19.6$ ms^{-1} //

(Total 5 marks)